S.NO	TASK	SUBTASK	STATUS	UPDATE DATE	DESCRIPTION	Duration
1	Lane Following					
1.1		Detect lane markings from camera frames	Completed	2/3/2026	BEV transform and thresholding for lane extraction.	
1.2		Track lanes using polynomial fitting	Completed	2/4/2026	Sliding window and polynomial lane tracking.	
1.3		Reduce lane wobble	Completed	2/5/2026	Applied smoothing for stable lane center.	
1.4		Integrate lane detection with steering control	Completed	2/7/2026	Stanley controller for lane keeping.	12 days
1.5		Handle temporary lane detection loss	Completed	2/8/2026	Maintain stable motion during short lane loss.	
1.6		Adjust speed based on lane curvature	Completed	2/9/2026	Reduce speed for curves and unstable lanes.	
1.7		Validate lane following on BFMC track	Completed	2/12/2026	Verified stable lane tracking on track runs.	
1.8		Optimize pipeline for real-time execution	Completed	2/14/2026	Improved processing speed for smooth control.	
2	TASK	SUBTASK	STATUS	UPDATE DATE	DESCRIPTION	Duration
	Dataset Creation					
2.1	Data Collection	Capture lane images under varied conditions	Completed	2/3/2026	Collected lane images from BFMC track under varied lighting and road conditions.	
2.2	Data Cleaning	Remove low-quality and duplicate images	Completed	2/4/2026	Removed low-quality and duplicate frames to ensure reliable training data.	
2.3	Annotation	Create and verify lane labels	Completed	2/5/2026	Created accurate lane labels/masks for supervised training.	6 days
2.4	Dataset Split	Split data into train, validation, and test sets	Completed	2/7/2026	Split dataset into training, validation, and testing sets.	
2.5	Preprocessing	Apply resizing, normalization, and augmentation	Completed	2/8/2026	Applied resizing, normalization, and basic augmentation for robustness.	
3	TASK	SUBTASK	STATUS	UPDATE DATE	DESCRIPTION	Duration
	Model Training					
3.1	Architecture Selection	Select lightweight CNN model	Completed	2/9/2026	Selected lightweight CNN suitable for real-time inference.	
3.2	Training Configuration	Configure optimizer and training parameters	Completed	2/11/2026	Configured optimizer, loss function, and training parameters.	
3.3	Model Training	Train model and save best weights	Completed	2/12/2026	Trained model and monitored training/validation performance.	7 days
3.4	Evaluation	Evaluate performance on test dataset	Completed	2/14/2026	Validated model performance on unseen test data.	
3.5	Optimization	Optimize model for real-time inference	Completed	2/15/2026	Optimized model for faster inference on Raspberry Pi.	
4	TASK	SUBTASK	STATUS	UPDATE DATE	DESCRIPTION	Duration
4.1	Yolo Integration	Model loading and initialization	Completed	2/16/2026	YOLO model integrated into perception pipeline.	
4.2		Real-time camera inference	Completed	2/17/2026	Implemented real-time object detection from camera feed.	
4.3		Detection filtering	Completed	2/18/2026	Applied confidence threshold to reduce false detections.	5 days
4.4		Behavior trigger mapping	Completed	2/19/2026	Linked detection outputs to vehicle actions.	
4.5		Real-track performance testing	Completed	2/20/2026	Evaluating detection stability during full-track runs.	
5	TASK	SUBTASK	STATUS	UPDATE DATE	DESCRIPTION	Duration
5.1	FSM integration	State machine framework setup	Completed	2/21/2026	Implemented FSM for vehicle behavior control.	
5.2		Lane following state	Completed	2/22/2026	Default driving state for lane tracking.	
5.3		Stop / crosswalk state	Completed	2/23/2026	Vehicle stops based on detected traffic signs.	
5.4		Parking state	Completed	2/24/2026	FSM transition for parking maneuver execution.	6 days
5.5		Pedestrian handling state	Completed	2/25/2026	Stops vehicle when pedestrian detected.	
5.6		State transition testing	Completed	2/26/2026	Validated smooth transitions between states.	
6	TASK	SUBTASK	STATUS	UPDATE DATE	DESCRIPTION	Duration
6.1	System Integration	Integrate perception, control, behavior modules	Completed	2/28/2026	Integrated full autonomous pipeline.	2 days
7	TASK	SUBTASK	STATUS	UPDATE DATE	DESCRIPTION	Duration
7.1	Autonomous Driving Validation	Validate full autonomous driving pipeline	Completed	3/1/2026	Verified stable autonomous driving.	6 days
8	TASK	SUBTASK	STATUS	UPDATE DATE	DESCRIPTION	Duration
8.1	Traffic Rule Testing	Test traffic rule behaviors	Completed	03/02/2026	Tested stop, crosswalk, and signals.	4 days
9	TASK	SUBTASK	STATUS	UPDATE DATE	DESCRIPTION	Duration
9.1	Parking Maneuver Testing	Validate parking FSM behavior	Completed	03/03/2026	Verified parking maneuver execution.	2 days
10	TASK	SUBTASK	STATUS	UPDATE DATE	DESCRIPTION	Duration
10.1	Real-Time Optimization	Optimize system performance	Completed	03/04/2026	Improved real-time processing.	4 days
11	TASK	SUBTASK	STATUS	UPDATE DATE	DESCRIPTION	Duration
11.1	Dashboard Development	Develop monitoring dashboard	Completed	03/05/2026	Implemented telemetry interface.	4 days
12	TASK	SUBTASK	STATUS	UPDATE DATE	DESCRIPTION	Duration
12.1	Robustness Testing	Perform system stress testing	Completed	03/06/2026	Tested system stability.	6 days
13	TASK	SUBTASK	STATUS	UPDATE DATE	DESCRIPTION	Duration
13.1	Qualification Video Preparation	Prepare BFMC demo video	Completed	03/07/2026	Recording autonomous runs.	1 day